

Home Work 5
II B. Tech. (Common)
Mathematics-II (NMCI102)

Notations:

$M_{m \times n}(\mathbb{F})$: The set of all matrices of order $m \times n$ with entries from $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$.

$M_n(\mathbb{F})$: The set of all matrices of order $n \times n$ with entries from $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$.

$P(\mathbb{R})$ or $\mathbb{R}[x]$: The set of all polynomials in one variable with real coefficients.

$P_n(\mathbb{R})$ or $\mathbb{R}_n[x]$: The set of all polynomials of degree at most n in one variable with real coefficients.

1. Examine whether T is a linear transformation. If T is linear, find $\text{Ker}(T)$ and $\text{Im}(T)$, verify the rank-nullity theorem for T , and determine whether T is one-to-one, onto, or both.
 - (a) $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $T(a_1, a_2) = (a_1, -a_2)$ (b) $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $T(a_1, a_2) = (a_1, 0)$
 - (c) $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ defined by $T(x, y) = (x + 2y, 2x + y, x + 2y)$ (d) $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ defined by $T(x, y, z) = (x - y, 2z)$
 - (e) $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by $T(x, y, z) = (yz, zx, xy)$ (f) $T: M_{m \times n}(\mathbb{R}) \rightarrow M_{n \times m}(\mathbb{R})$ defined by $T(A) = A^t$
 - (g) $T: M_{n \times n}(\mathbb{R}) \rightarrow M_{n \times n}(\mathbb{R})$ defined by $T(A) = \frac{1}{2}(A + A^t)$ (h) $T: P_n(\mathbb{R}) \rightarrow P_{n-1}(\mathbb{R})$ defined by $T(f(x)) = f'(x)$
 - (i) $T: P_2(\mathbb{R}) \rightarrow \mathbb{R}$ defined by $T(f(x)) = \int_a^b f(t)dt$, where $a, b \in \mathbb{R}$ and $a < b$
 - (j) $T: P_2(\mathbb{R}) \rightarrow \mathbb{R}^3$ defined by $T(a_0 + a_1x + a_2x^2) = (a_0, a_1, a_2)$
 - (k) $T: P_2(\mathbb{R}) \rightarrow P_3(\mathbb{R})$ defined by $T(f(x)) = 2f'(x) + \int_0^x 3f(t)dt$
 - (l) $T: P_2(\mathbb{R}) \rightarrow M_{2 \times 2}(\mathbb{R})$ defined by $T(f(x)) = \begin{bmatrix} f(1) - f(2) & 0 \\ 0 & f(0) \end{bmatrix}$
 - (m) $T: M_{2 \times 2}(\mathbb{R}) \rightarrow \mathbb{R}$ defined by $T(A) = \text{trace}(A)$
 - (n) $T: P_2(\mathbb{R}) \rightarrow P_3(\mathbb{R})$ defined by $T(f(x)) = \int_0^x f(t)dt$
 - (o) $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $T(a_1, a_2) = (1, a_2)$ (p) $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $T(a_1, a_2) = (a_1, a_1^2)$
 - (q) $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $T(a_1, a_2) = (\sin a_1, 0)$ (r) $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $T(a_1, a_2) = (|a_1|, a_2)$
 - (s) $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $T(a_1, a_2) = (a_1 + 1, a_2)$.

Answers: (a) Yes, $\text{Ker}(T) = \{(0, 0)\}$, $\text{Im}(T) = \mathbb{R}^2$ (b) Yes, $\text{Ker}(T) = Y - \text{axis}$, $\text{Im}(T) = X - \text{axis}$ (c) Yes, $\text{Ker}(T) = \{(0, 0)\}$, $\text{Im}(T) = L\{(1, 0, 1), (0, 1, 0)\}$ (d) Yes, $\text{Ker}(T) = L\{(1, 1, 0)\}$, $\text{Im}(T) = \mathbb{R}^2$ (e) No (f) Yes, $\text{Ker}(T) = 0_{m \times n}$, $\text{Im}(T) = M_{n \times m}(\mathbb{R})$ (g) Yes, $\text{Ker}(T) = \{A \in M_{n \times n}(\mathbb{R}) : A = A^t\}$, $\text{Im}(T) = \{A \in M_{n \times n}(\mathbb{R}) : A = -A^t\}$ (h) Yes, $\text{Ker}(T) = \{\text{all constant polynomials in } P_n(\mathbb{R})\}$, $\text{Im}(T) = P_{n-1}(\mathbb{R})$ (i) Yes (j) Yes (k) Yes (l) Yes (m) Yes (n) Yes (o) No (p) No (q) No (r) No (s) No.

2. (a) Is there a linear transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ such that $T(1, 0) = (1, 4)$ and $T(1, 1) = (2, 5)$? If yes, what is $T(2, 3)$?
(b) Is there a linear transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ such that $T(1, 1) = (1, 0, 2)$ and $T(2, 3) = (1, -1, 4)$? If yes, find T and what is $T(8, 11)$?
(c) Is there a linear transformation $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ such that $T(1, 0, 3) = (1, 1)$ and $T(-2, 0, -6) = (2, 1)$?
(d) Determine the linear transformation $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ which maps the basis vectors $(0, 1, 1)$, $(1, 0, 1)$, $(1, 1, 0)$ of $\mathbb{R}^3$ to $(1, 1, 1)$, $(1, 1, 1)$, $(1, 1, 1)$ respectively. Verify rank-nullity theorem after finding $\text{Ker}(T)$, $\text{Im}(T)$.

Answers: (a) Yes, $(5, 11)$ (b) Yes, $(5, -3, 16)$ (c) No (d) $T(x, y, z) = \left(\frac{x+y+z}{2}, \frac{x+y+z}{2}, \frac{x+y+z}{2} \right)$.

3. Find all linear transformations $T: \mathbb{F} \rightarrow \mathbb{F}$ where $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$.
4. Find all linear transformations $T: \mathbb{F}^2 \rightarrow \mathbb{F}^2$ where $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$.

5. A linear transformation $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ is defined by $T(x_1, x_2, x_3) = (3x_1 - 2x_2 + x_3, x_1 - 3x_2 - 2x_3)$. Find the matrix of T relative to the ordered bases
- (a) $\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ of $\mathbb{R}^3$ and $\{(1, 0), (0, 1)\}$ of $\mathbb{R}^2$.
- (b) $\{(0, 1, 0), (1, 0, 0), (0, 0, 1)\}$ of $\mathbb{R}^3$ and $\{(0, 1), (1, 0)\}$ of $\mathbb{R}^2$.
- (c) $\{(0, 1, 1), (1, 0, 1), (1, 1, 0)\}$ of $\mathbb{R}^3$ and $\{(1, 0), (0, 1)\}$ of $\mathbb{R}^2$.

Answers: (a) $\begin{bmatrix} 3 & -2 & 1 \\ 1 & -3 & -2 \end{bmatrix}$ (b) $\begin{bmatrix} -3 & 1 & -2 \\ -2 & 3 & 1 \end{bmatrix}$ (c) $\begin{bmatrix} -1 & 4 & 1 \\ -5 & -1 & -2 \end{bmatrix}$.

6. The matrix of a linear transformation $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ with respect to the ordered basis $\{(0, 1, 1), (1, 0, 1), (1, 1, 0)\}$ of $\mathbb{R}^3$ is given by $\begin{bmatrix} 0 & 3 & 0 \\ 2 & 3 & -2 \\ 2 & -1 & 2 \end{bmatrix}$. Find T and also find the matrix of T with respect to the ordered basis $\{(2, 1, 1), (1, 2, 1), (1, 1, 2)\}$ of $\mathbb{R}^3$.

Answers: $T(x, y, z) = (-x + y + 3z, x + y + z, x - 3y + 5z)$, $m(T) = \begin{bmatrix} -1/2 & 2 & 3/2 \\ 3/2 & 2 & -1/2 \\ 3/2 & -2 & 7/2 \end{bmatrix}$.